Assignment D1-V1: Planetary Rover – Single Rover, Static Environment

Scenario

A single wheeled rover operates on a planetary surface represented as a finite graph of locations. The rover starts at a base station and must visit a set of target locations to collect geological samples.

Each movement between locations consumes energy. The rover has a limited battery capacity and cannot exceed it. Samples are collected at specific locations and must be returned to the base station.

The environment is static (no obstacles appear or disappear), and all locations are known in advance.

Domain Characteristics

- Robot: single wheeled rover
- Environment: static graph of locations
- Resources: battery level
- Tasks: navigation, sample collection, sample delivery
- Constraints: limited energy capacity

Modelling Guidelines

- Clearly distinguish between navigation, sampling, and delivery actions.
- Model battery as a numeric fluent.
- Avoid trivial encodings (e.g., directly encoding goal satisfaction in actions).
- Ensure that energy constraints meaningfully affect plan feasibility.

Q1 – Basic PDDL Model

Model the domain using classical PDDL with numeric fluents.

You must:

- Define appropriate predicates and functions.
- Provide at least two problem instances:
  - one simple instance (few locations, single sample)
  - one non-trivial instance (multiple samples, constrained battery)
- Provide a valid plan for each problem.
- Briefly justify your modelling choices (especially energy representation).

Q2 – PDDL+ Model

Extend the same domain using PDDL+.

You must:

- Introduce a continuous battery drain process (even when idle).
- Introduce an event that triggers when battery reaches a critical threshold.
- Explain how the system evolves autonomously over time.
- Provide at least two problem instances where:
  - idle time affects feasibility
  - the timing of actions matters

Discussion

Discuss:

- differences between discrete and continuous energy modelling
- limitations of your abstraction